A project
report on

Privacy Aware Intelligent Smart IoT Sensor System

Submitted for partial fulfillment of the requirements for the
Internet of Things subject

of
Bachelor of Engineering

by

1. Zishan Shaikh TAI&DB01
2. Ashish Maurya TAI&DB04

Under the guidance of
Prof. Siddhesh Gadge

DYP DPU
Dr. D. Y. Patil Institute of Technology, Pimpri, Pune

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Academic Year 2026-2027

DYP DPU
Dr. D. Y. Patil Institute of Technology, Pimpri, Pune

CERTIFICATE

This is to certify that the project work entitled **Privacy Aware Intelligent Smart IoT Sensor System** is a bona fide work carried out by **Zishan Shaikh (TAI&DB01;)** and **Ashish Maurya (TAI&DB04;)** in partial fulfillment of the requirements for the subject **Internet of Things** of Bachelor of Engineering during the academic year 2026-2027.

Prof. Siddhesh Gadge	**Head of Department**
Project Guide	**AI and DS**

Acknowledgement

We express our sincere gratitude to our Project Guide, **Prof. Siddhesh Gadge**, for his guidance, encouragement and constructive suggestions throughout the development of this IoT project. His support helped us translate the idea of privacy-aware sensing into a working simulation.

We thank the Department of Artificial Intelligence and Data Science and Dr. D. Y. Patil Institute of Technology, Pimpri, Pune for providing the learning environment and resources required for this work. We also thank everyone who supported the planning, design, implementation and evaluation of the project.

Zishan Shaikh and Ashish Maurya

Abstract

Smart sensor deployments can improve safety and resource use in buildings, but uncontrolled collection can expose device identity, detailed location and behavioural context. This project presents Sentinel AIoT, a privacy-aware intelligent smart sensor system that simulates temperature, humidity, carbon dioxide and light sensing in a connected building.

The system models an end-to-end pipeline: edge devices generate readings, a secure gateway verifies devices and messages, a privacy filter pseudonymises the device session and reduces location to an approved zone, and a cloud preprocessing service creates quality-checked feature vectors. An Isolation Forest style anomaly score is combined with policy rules to recommend safe actions such as increasing fresh-air intake or enabling cooling. The project also provides a multi-page interactive console for operations, analytics, security events and project guidance.

Keywords: Internet of Things, AIoT, smart sensors, privacy by design, TLS, mutual authentication, cloud preprocessing, anomaly detection.

Table of Contents

Chapter	Topic
I	Introduction
II	Existing System
III	Problem Identification
IV	Methodology
V	System Design
VI	Implementation
VII	Conclusion and Future Scope
VIII	References

List of Figures

Figure 1: Sentinel AIoT end-to-end processing pipeline
Figure 2: Smart building digital twin and virtual sensor placement
Figure 3: Privacy-aware cloud preprocessing flow
Figure 4: Multi-page operations, analytics and security console

List of Acronyms

Acronym	Meaning
AIoT	Artificial Intelligence of Things
TLS	Transport Layer Security
mTLS	Mutual Transport Layer Security
MQTT	Message Queuing Telemetry Transport
CO2	Carbon Dioxide
ML	Machine Learning

Chapter I: Introduction

1.1 Domain

The Internet of Things connects physical devices that sense, communicate and act. AIoT extends this idea with machine learning so that sensor data can be converted into timely recommendations. This project is situated in smart-building environmental monitoring, where indoor conditions affect comfort, ventilation and energy consumption.

1.2 Objectives

• Design a working simulation for multiple environmental sensors and a secure gateway.

• Apply privacy-by-design controls before cloud analytics.

• Perform cloud-style preprocessing and anomaly-oriented decision making.

• Demonstrate normal, poor-air-quality and heat-anomaly scenarios.

• Provide operations, analytics and security views suitable for an academic demonstration.

1.3 Problem Specification

A conventional smart sensor system may forward all raw messages to the cloud and identify a device by a persistent identifier. Such a design creates unnecessary privacy and security risk. The problem is to make useful ventilation and cooling decisions while collecting only the data necessary for that purpose.

Chapter II: Existing System

Existing smart monitoring systems often focus on data capture and visual dashboards. They may lack end-to-end device identity verification, explainable processing steps, bounded retention and privacy-aware handling of contextual data. Alerting can also be based on a single threshold, which may produce false alarms when values fluctuate normally.

Limitations of Existing Systems

• Persistent device identifiers may be stored in analytical datasets.

• Exact location may be retained even when zone-level context is sufficient.

• Raw data may be retained without an operational retention limit.

• Anomaly decisions may not expose confidence, policy logic or operator override.

• Security, operations and analytics evidence may be shown in separate tools.

Proposed Improvement

Sentinel AIoT separates sensing, secure transport, privacy filtering, feature processing and decision policy. It also exposes these stages in one demonstrator, making the system easier to explain and evaluate.

Chapter III: Problem Identification

3.1 Problem Definition

Develop a privacy-aware intelligent smart IoT system that receives environmental sensor data, verifies secure communication, preprocesses data in a cloud-style pipeline and produces a machine-learning-assisted decision without retaining unnecessary identifying information.

3.2 Functional Requirements

Requirement	System response
Sensor simulation	Generate temperature, humidity, CO2 and light telemetry.
Secure ingestion	Model TLS 1.3, mutual device authentication and topic authorization.
Privacy controls	Use pseudonymous sessions, zone-level location and retention policy.
Decision support	Calculate anomaly score and provide action recommendation.
Observability	Show telemetry, analytics, device registry and security audit streams.

3.3 Non Functional Requirements

The project should be understandable, responsive, modular, auditable and safe for demonstration. Real equipment is not controlled; generated readings are synthetic and used only to represent the system behaviour.

Chapter IV: Methodology

4.1 System Methodology

Each virtual sensor produces a measurement. The gateway validates the message format and device identity. The privacy filter transforms the message by replacing the device identifier with a session pseudonym and retaining only an approved zone. The preprocessing layer removes duplicates, checks ranges and derives rolling features. The model scores deviation from learned normal behaviour, and a policy rule turns the score into an advisory.

4.2 Algorithm Used

Isolation Forest style anomaly detection: Normal behaviour is represented by a feature vector containing values such as temperature, CO2 trend, rolling mean and variance. A high anomaly score indicates that the current vector is unlike normal operating data. The prototype uses a review threshold of 0.72. The final decision combines the score with data-quality and safety-policy checks.

4.3 Software and Hardware Requirements

Category	Requirement
Software	HTML, CSS, JavaScript, modern web browser
Simulation devices	Four virtual sensors and one gateway
Security concepts	TLS 1.3, mutual TLS, access control list, certificate rotation
Cloud concepts	Queue, preprocessing service, feature store, inference service
Machine learning	Isolation Forest anomaly-detection approach

System Design

5.1 Architecture

The architecture follows a layered pipeline: Edge Sensors -> Secure Gateway -> Privacy Filter -> Cloud Feature Service -> ML Decision Engine -> Operations and Security Console. The gateway performs early validation so invalid messages are rejected before persistence. The privacy filter minimises the record and the decision engine preserves a human override.

5.2 Data Flow

Stage	Input	Output
Edge	Environmental reading	Validated device payload
Gateway	Payload and certificate	Authenticated permitted message
Privacy filter	Device and contextual metadata	Pseudonymous zone-level record
Preprocessing	Accepted record	Cleaned feature vector
Decision engine	Feature vector	Score, classification and recommendation

5.3 Privacy and Security Design

• TLS 1.3 and mutual TLS model encrypted authenticated transport.

• Device identifiers are replaced with session pseudonyms before analytics.

• Precise location is reduced to approved building zones.

• Raw messages use a bounded 24-hour retention policy in the design.

• Operations, security and model roles receive only the minimum information needed.

Chapter VI: Implementation

6.1 Multi Page Simulation

The implementation is a browser-based project console. The Operations Lab renders a smart-building digital twin, live sensors, chart, message stream and ML decision panel. Users can switch among normal operation, poor air quality and heat anomaly scenarios. The Analytics Centre displays feature-vector flow, data quality, drift and model service health. The Security Centre displays device registry, certificate status, control posture and a security event stream.

6.2 Scenario Behaviour

Scenario	Observed condition	System decision
Normal operation	Measurements remain inside baseline.	Maintain normal ventilation.
Poor air quality	CO2 trend becomes abnormal.	Recommend increasing fresh-air intake and request review.
Heat anomaly	Temperature departs from baseline.	Recommend cooling and notify operator.

6.3 Testing

Test	Expected Result	Result
Telemetry stream	Packets and pseudonymous records update.	Pass
Air quality scenario	Alert and fresh-air recommendation appear.	Pass
Heat scenario	Temperature alert and cooling recommendation appear.	Pass
Analytics view	Feature and model streams update.	Pass
Security view	Registry and audit events update.	Pass

Chapter VII: Conclusion and Future Scope

7.1 Conclusion

The Sentinel AIoT project demonstrates that a smart sensor application can provide useful operational decisions while applying privacy and security controls throughout the pipeline. The system presents a clear path from simulated sensing to authenticated transport, data minimisation, cloud preprocessing, anomaly scoring and operator-visible action. The multi-page console makes each system responsibility visible for academic evaluation.

7.2 Advantages

• Combines sensing, secure communication, privacy and ML decision support.

• Explains data handling through visible operations, analytics and security streams.

• Supports multiple realistic scenario demonstrations.

• Uses bounded and privacy-safe analytical representation.

• Keeps human override visible for consequential actions.

7.3 Limitations and Future Scope

The project uses synthetic browser-based telemetry and does not connect to physical hardware. Future work can integrate ESP32 or Raspberry Pi sensor nodes, an MQTT broker, managed public-key infrastructure, cloud database, real model training pipeline, role-based login, encrypted key management and model-drift monitoring. A site-specific threat model and data protection impact assessment should precede a real deployment.

References

[1] OASIS. MQTT Version 5.0 Specification.

[2] NIST. Privacy Framework: A Tool for Improving Privacy Through Enterprise Risk Management.

[3] NIST. Cybersecurity Framework 2.0.

[4] scikit-learn documentation. IsolationForest for anomaly detection.

[5] OWASP. Internet of Things Security Guidance.

[6] IETF. The Transport Layer Security Protocol Version 1.3, RFC 8446.